DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

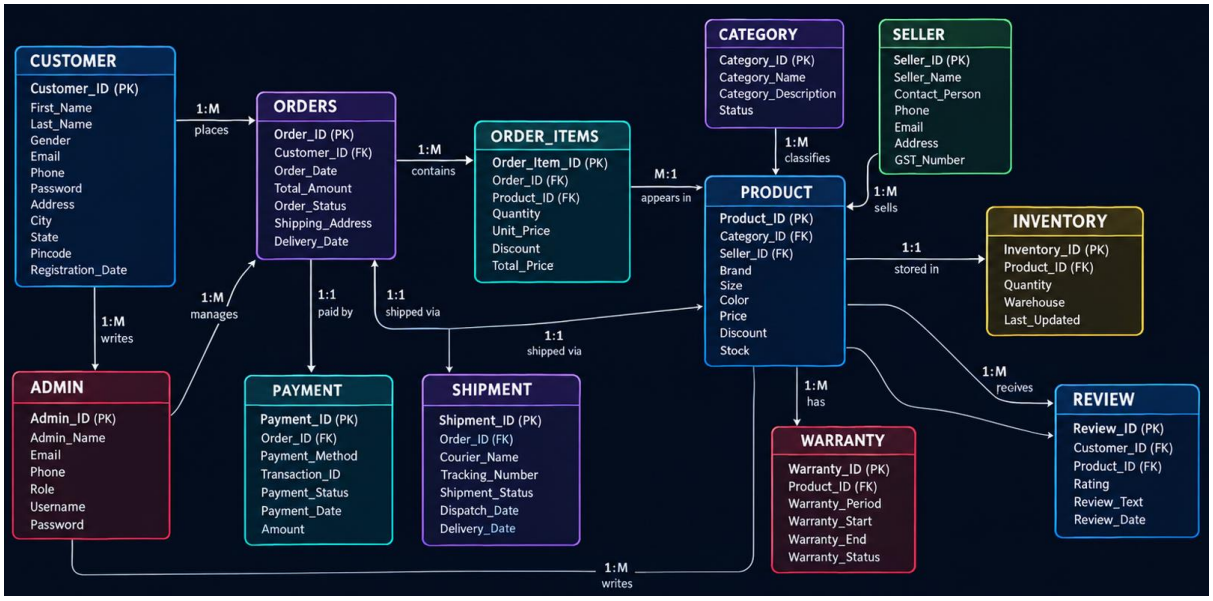
Project Title: Myntra – E Commerce Order Management System Skill

Objectives

1. Combine Customer, Product, Order, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

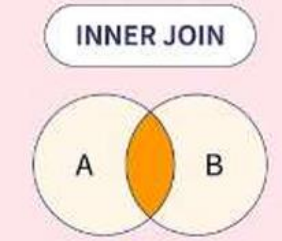
1.Combining Multiple Tables

The myntra database contains different tables that store customer, product, order, and payment information. The tables can be combined using JOIN operations to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> -- 2.1 INNER JOIN: Customer and Orders
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID, o.Order_Date, o.Total_Amount
  4  FROM Customer c
  5  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- -------------------- ---------- --------- ------------
        101 Rahul Sharma                501 10-SEP-26      1231.12
        102 Priya Ravi                  502 11-SEP-26       3398.3
        103 Karthik Kumar               503 12-SEP-26       2799.2
        104 Anitha Suresh               504 13-SEP-26      5699.05
        105 Arun Prakash                505 14-SEP-26       1619.1
        106 Divya Lakshmi               506 15-SEP-26      2462.24

6 rows selected.
```

2.2 Orders and Payment

```
SQL> SELECT o.Order_ID, o.Order_Date, o.Total_Amount,
  2         p.Payment_Method, p.Amount AS Payment_Amount, p.Payment_Status
  3  FROM Orders o
  4  INNER JOIN Payment p ON o.Order_ID = p.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_METHOD  PAYMENT_AMOUNT PAYMENT_STAT
---------- --------- ------------ --------------- -------------- ------------
       501 10-SEP-26      1231.12 UPI                      1300 Successful
       503 12-SEP-26       2799.2 UPI                    2799.2 Successful
       502 11-SEP-26       3398.3 Card                   3398.3 Successful
       504 13-SEP-26      5699.05 Card                  5699.05 Successful
       505 14-SEP-26       1619.1 COD                    1619.1 Successful
       506 15-SEP-26      2462.24 UPI                   2462.24 Successful

6 rows selected.
```

2.3 Product and Order_Items

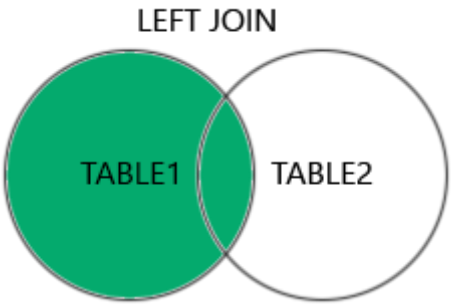
```
SQL> SELECT pr.Product_ID, pr.Product_Name, oi.Order_ID,
  2         oi.Quantity, pr.Price AS Unit_Price,
  3         oi.Quantity * pr.Price AS Subtotal
  4  FROM Product pr
  5  INNER JOIN Order_Items oi ON pr.Product_ID = oi.Product_ID;

PRODUCT_ID PRODUCT_NAME          ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- -------------------- ---------- ---------- ---------- ----------
       201 Cotton Shirt                501          1       1399       1399
       202 Floral Dress                502          2       1999       3998
       203 Running Shoes               503          1       3499       3499
       204 Analog Watch                504          1       5999       5999
       205 Backpack                    505          1       1799       1799
       201 Cotton Shirt                506          2       1399       2798

6 rows selected.
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

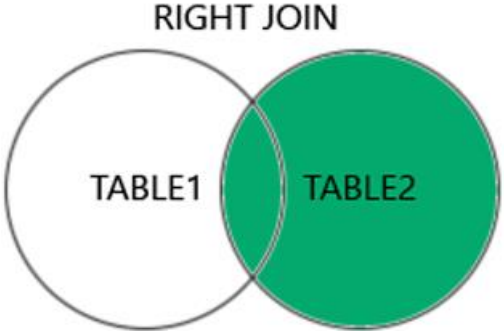
```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID, o.Order_Date, o.Total_Amount
  4  FROM Customer c
  5  LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- -------------------- ---------- --------- ------------
        101 Rahul Sharma                501 10-SEP-26      1231.12
        102 Priya Ravi                  502 11-SEP-26       3398.3
        103 Karthik Kumar               503 12-SEP-26       2799.2
        104 Anitha Suresh               504 13-SEP-26      5699.05
        105 Arun Prakash                505 14-SEP-26       1619.1
        106 Divya Lakshmi               506 15-SEP-26      2462.24
        107 Vignesh Raj
        108 Meena Krishnan

8 rows selected.
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID, o.Order_Date, o.Total_Amount
  4  FROM Customer c
  5  RIGHT JOIN Orders o ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------- -------------------- ---------- --------- ------------
        101 Rahul Sharma                501 10-SEP-26      1231.12
        102 Priya Ravi                  502 11-SEP-26       3398.3
        103 Karthik Kumar               503 12-SEP-26       2799.2
        104 Anitha Suresh               504 13-SEP-26      5699.05
        105 Arun Prakash                505 14-SEP-26       1619.1
        106 Divya Lakshmi               506 15-SEP-26      2462.24

6 rows selected.
```

5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID, o.Order_Date,
  4         pr.Product_ID, pr.Product_Name,
  5         oi.Quantity, pr.Price AS Unit_Price,
  6         oi.Quantity * pr.Price AS Subtotal,
  7         py.Payment_Method, py.Amount AS Payment_Amount, py.Payment_Status
  8  FROM Customer c
  9  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
 10  INNER JOIN Order_Items oi ON o.Order_ID = oi.Order_ID
 11  INNER JOIN Product pr ON oi.Product_ID = pr.Product_ID
 12  INNER JOIN Payment py ON o.Order_ID = py.Order_ID
 13  ORDER BY o.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DAT PRODUCT_ID PRODUCT_NAME         QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_METHOD  PAYMENT_AMOUNT PAYMENT_ST
AT
----------- -------------------- ---------- --------- ---------- -------------------- ---------- ---------- ---------- --------------- -------------- ----------
--
        106 Divya Lakshmi               506 15-SEP-26        201 Cotton Shirt                2       1399       2798 UPI                2462.24 Successful
        105 Arun Prakash                505 14-SEP-26        205 Backpack                    1       1799       1799 COD                 1619.1 Successful
        104 Anitha Suresh               504 13-SEP-26        204 Analog Watch                1       5999       5999 Card               5699.05 Successful
        103 Karthik Kumar               503 12-SEP-26        203 Running Shoes               1       3499       3499 UPI                 2799.2 Successful
        102 Priya Ravi                  502 11-SEP-26        202 Floral Dress                2       1999       3998 Card                3398.3 Successful
        101 Rahul Sharma                501 10-SEP-26        201 Cotton Shirt                1       1399       1399 UPI                1300 Successful

6 rows selected.

SQL>
```

6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT c.Customer_ID, c.First_Name || ' ' || c.Last_Name AS Customer_Name, o.Order_ID, o.Order_Date, pr.Product_Name, oi.Quantity, oi.Quantity * pr.Price AS Subtotal FROM Customer c I
NNER JOIN Orders o ON c.Customer_ID = o.Customer_ID INNER JOIN Order_Items oi ON o.Order_ID = oi.Order_ID INNER JOIN Product pr ON oi.Product_ID = pr.Product_ID ORDER BY c.Customer_ID, o.O
rder_Date;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DAT PRODUCT_NAME         QUANTITY   SUBTOTAL
----------- ------------------- ---------- --------- -------------------- ---------- ----------
        101 Rahul Sharma               501 10-SEP-26 Cotton Shirt                  1       1399
        102 Priya Ravi                 502 11-SEP-26 Floral Dress                  2       3998
        103 Karthik Kumar              503 12-SEP-26 Running Shoes                 1       3499
        104 Anitha Suresh              504 13-SEP-26 Analog Watch                  1       5999
        105 Arun Prakash               505 14-SEP-26 Backpack                      1       1799
        106 Divya Lakshmi              506 15-SEP-26 Cotton Shirt                  2       2798

6 rows selected.
```

7. Customer Purchase History with Payment

This query displays the products purchased by each customer along with payment mode and payment status.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID, o.Order_Date,
  4         pr.Product_Name, oi.Quantity,
  5         oi.Quantity * pr.Price AS Subtotal,
  6         py.Payment_Method, py.Payment_Status
  7  FROM Customer c
  8  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  9  INNER JOIN Order_Items oi ON o.Order_ID = oi.Order_ID
 10  INNER JOIN Product pr ON oi.Product_ID = pr.Product_ID
 11  INNER JOIN Payment py ON o.Order_ID = py.Order_ID
 12  ORDER BY c.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DAT PRODUCT_NAME         QUANTITY   SUBTOTAL PAYMENT_METHOD  PAYMENT_STAT
----------- ------------------- ---------- --------- -------------------- ---------- ---------- --------------- ------------
        101 Rahul Sharma               501 10-SEP-26 Cotton Shirt                  1       1399 UPI             Successful
        102 Priya Ravi                 502 11-SEP-26 Floral Dress                  2       3998 Card            Successful
        103 Karthik Kumar              503 12-SEP-26 Running Shoes                 1       3499 UPI             Successful
        104 Anitha Suresh              504 13-SEP-26 Analog Watch                  1       5999 Card            Successful
        105 Arun Prakash               505 14-SEP-26 Backpack                      1       1799 COD             Successful
        106 Divya Lakshmi              506 15-SEP-26 Cotton Shirt                  2       2798 UPI             Successful

6 rows selected.
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report showing total orders, total purchase amount, and total products for each customer.

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  3         SUM(oi.Quantity * pr.Price) AS Total_Purchase,
  4         SUM(oi.Quantity) AS Total_Products
  5  FROM Customer c
  6  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  7  INNER JOIN Order_Items oi ON o.Order_ID = oi.Order_ID
  8  INNER JOIN Product pr ON oi.Product_ID = pr.Product_ID
  9  GROUP BY c.First_Name, c.Last_Name
 10  ORDER BY Total_Purchase DESC;

CUSTOMER_NAME        TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
-------------------- ------------ -------------- --------------
Anitha Suresh                   1           5999              1
Priya Ravi                      1           3998              2
Karthik Kumar                   1           3499              1
Divya Lakshmi                   1           2798              2
Arun Prakash                    1           1799              1
Rahul Sharma                    1           1399              1

6 rows selected
```

9. Payment Transaction Report

This query displays each customer's orders along with payment method, payment amount, and payment status.

```
SQL>
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         o.Order_ID, o.Order_Date, o.Total_Amount,
  3         py.Payment_Method, py.Amount AS Payment_Amount, py.Payment_Status
  4  FROM Customer c
  5  INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID
  6  INNER JOIN Payment py ON o.Order_ID = py.Order_ID
  7  ORDER BY o.Order_Date DESC;

CUSTOMER_NAME         ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_METHOD  PAYMENT_AMOUNT PAYMENT_STAT
-------------------- ---------- --------- ------------ --------------- -------------- ------------
Divya Lakshmi               506 15-SEP-26      2462.24 UPI                    2462.24 Successful
Arun Prakash                505 14-SEP-26       1619.1 COD                     1619.1 Successful
Anitha Suresh               504 13-SEP-26      5699.05 Card                   5699.05 Successful
Karthik Kumar               503 12-SEP-26       2799.2 UPI                     2799.2 Successful
Priya Ravi                  502 11-SEP-26       3398.3 Card                    3398.3 Successful
Rahul Sharma                501 10-SEP-26      1231.12 UPI                       1300 Successful

6 rows selected.
```

Conclusion

In this task, INNER JOIN, LEFT JOIN, and RIGHT JOIN were used to combine the Customer, Orders, Order_Items, Product, and Payment tables. These joins helped retrieve complete order details, customer purchase history, and multi-table business reports.